

q-Weibull Competing Risks under Weak Identifiability: Bayesian Inference via the No-U-Turn Sampler

Huirong Cao¹  | Fuchang Wang² 

¹School of Science, Langfang Normal University, Langfang, China | ²University of Emergency Management, Sanhe, China |

Correspondence: Huirong Cao (huirongcao@126.com)

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ABSTRACT

Competing-risks data, where each unit fails from one of several mutually exclusive causes, arise routinely in reliability and lifetime analysis. The q-Weibull distribution nests the ordinary Weibull and accommodates both bounded ($q < 1$) and heavy-tailed ($q > 1$) lifetimes, but Bayesian inference for it under competing risks remains under-developed. We develop a unified Bayesian framework computed by the No-U-Turn Sampler (NUTS) in a better-conditioned, unconstrained coordinate system $(q, \log \eta, \log \beta)$, with an analytic gradient, Hessian-based preconditioning, and a smooth barrier at the bounded support wall. Because early competing failures remove the tail observations from which q is learned, competing truncation leaves q weakly identifiable: the profile likelihood is broad and Wald intervals under-cover. Across six shape values spanning the bounded, Weibull, and heavy-tailed regimes and three truncation levels, we show that, *specifically under strong competing-risk truncation and when the true q is moderately close to the regularising centre*, a quadratic q penalty (prior shrinkage) reduces the root-mean-squared error and restores near-nominal coverage. A controlled comparison with unpenalised MLE, penalised MLE, and the mode (MAP) shows that most of this point-estimate gain is attributable to the regularisation itself rather than to Bayesian computation; the added value of the full Bayesian analysis is calibrated posterior intervals, whose coverage we document. When the tail is observed or q lies far from the prior centre, the likelihood dominates and the MLE is preferable. NUTS adapts its mass matrix automatically and uses gradient evaluations about twice as efficiently as fixed-trajectory HMC, and its effective-sample throughput is comparable to a mature probabilistic-programming implementation on the same data. On high-voltage stress-test data ($n = 58$), competing risks are significant, but the early mode shows only an uncertain heavy-tail signal, and posterior predictive checks are consistent with the observed lifetimes; the degradation mode is well described by an ordinary Weibull.

1 | Introduction

Competing risks occur when a unit is exposed to several mutually exclusive failure mechanisms and the observed lifetime is the minimum of the latent cause-specific times, together with an indicator of the cause responsible. Such data are routine in the analysis of electronic components, mechanical systems, and accelerated life tests [10, 18]. Because failure by one cause removes the unit from exposure to all others, the latent times of the non-operating causes are censored, a structure that must be respected by both the likelihood and any inferential procedure.

The Weibull distribution has long been the workhorse of parametric reliability analysis. Its fixed stretched-exponential tail, however, cannot represent the heavier-than-Weibull tails associated with wear-out heterogeneity, nor the finite upper support that arises for certain bounded-failure processes. The q-Weibull distribution, obtained

from Tsallis entropy [22] and studied in reliability by several authors [9, 15], nests the ordinary Weibull at $q = 1$, gives a heavy (power-law) tail for $1 < q < 2$, and a bounded support for $q < 1$. A single additional parameter thus extends the Weibull in two useful directions.

Despite this flexibility, inference for q-Weibull competing risks faces three difficulties. *First*, the parameter that governs the tail, q , is identified largely from observations in the tail; when a competing cause truncates the lifetime distribution early, the tail of the cause of interest is only partially observed and q can become weakly identifiable, with a profile likelihood that is broad, flat, or asymmetric. *Second*, maximum-likelihood estimates and their Wald standard errors rely on local quadratic and asymptotic-normal approximations that are unreliable in this limited-information regime, and the bounded regime introduces a hard support boundary at which the

log-density is singular. *Third*, generic random-walk Metropolis samplers mix poorly over the correlated, curved q-Weibull parameter space, making a naive Bayesian implementation computationally expensive.

This article addresses all three difficulties with a unified Bayesian treatment computed by the No-U-Turn Sampler (NUTS) [14], an adaptive form of Hamiltonian Monte Carlo (HMC) [17, 7] that uses gradient information and recursive trajectory doubling. Recent work has applied HMC to Weibull-based competing risks [2, 3, 1], used NUTS for Bayesian inference of short- and long-tailed queueing models [5], and applied Gibbs sampling to the q-Weibull [19]. The same high-voltage endurance data have been analysed with ordinary Weibull competing risks under adaptive progressive censoring [12]. The present framework differs by targeting the q-Weibull directly, by using NUTS rather than Gibbs or random-walk proposals, and by treating identifiability under competing truncation as a first-class question. Our contributions are:

1. We establish a **Bayesian framework for q-Weibull competing risks under partially observed tails**, formulated in an unconstrained parameterisation with substantially reduced global posterior anisotropy $(q, \log \eta, \log \beta)$, with a cause-separable likelihood and an analytic log-posterior gradient that remains valid continuously across $q = 1$.
2. We **characterise the loss of identifiability of q induced by competing-risk truncation**: early competing failures remove the tail observations from which q is learned, giving broad, flat, or asymmetric profile likelihoods, and we document how this affects the bounded and heavy-tailed regimes differently.
3. We develop a **computational strategy** combining an unconstrained parameterisation, analytic gradients, Hessian preconditioning, and a smooth treatment of the bounded-support wall and the parameter bounds, so that Hamiltonian trajectories explore the posterior without entering singular regions.
4. Through an **extensive simulation over the bounded, Weibull, and heavy-tailed regimes**, we show that Bayesian regularisation improves root-mean-squared error and interval coverage specifically in the weakly identified, strongly truncated regime, whereas the likelihood dominates when the tail is observed.
5. We give a **reliability application** that shows how the framework distinguishes genuine evidence for a heavy tail from a mere displacement of the posterior above $q = 1$, avoiding over-interpretation of a small, heavily censored sample.

The remainder of the article is organised as follows. Section 2 defines the q-Weibull distribution and its properties. Section 3 develops the competing-risks model, likelihood, and derived quantities. Section 4 presents the Bayesian formulation, and Section 5 the NUTS computation including the smooth barrier and diagnostics. Sections 6 and 7 report the simulation study and the high-voltage application, respectively. Section 8 discusses the findings, and Section 9 concludes.

2 | The q-Weibull distribution

A q-Weibull random variable $T > 0$ has survival function

$$R(t) = \Pr(T > t) = [1 - (1 - q) \lambda t^\beta]^{1/(1-q)}, \quad (1)$$

where $\beta > 0$ is a shape parameter, $\lambda > 0$ a scale-related parameter, and $q < 2$ the entropic shape parameter. Define the *inner factor*

$$u(t) := 1 - (1 - q) \lambda t^\beta. \quad (2)$$

The probability density and hazard functions are

$$f(t) = (2 - q) \beta \lambda t^{\beta-1} u(t)^{1/(1-q)}, \quad (3)$$

$$h(t) = \frac{(2 - q) \beta \lambda t^{\beta-1}}{u(t)}. \quad (4)$$

It is convenient to express the scale through the characteristic life $\eta > 0$, setting

$$\lambda = \eta^{-\beta}. \quad (5)$$

2.1 | Parameter regimes

Equation (1) contains three regimes.

- *Ordinary Weibull* ($q \rightarrow 1$). Taking the limit, $R(t) \rightarrow \exp(-\lambda t^\beta) = \exp[-(t/\eta)^\beta]$, the standard two-parameter Weibull survival function.
- *Heavy tail* ($1 < q < 2$). Here $1 - q < 0$, so $u(t) = 1 + (q - 1)\lambda t^\beta > 1$ for all t and the support is unbounded. The survival function decays as a power law, $R(t) \asymp t^{-\beta(2-q)/(q-1)}$, giving a tail heavier than the Weibull's stretched exponential.
- *Bounded support* ($q < 1$). Here $u(t)$ reaches zero at the finite upper endpoint

$$T_{\max} = \left[\frac{1}{(1 - q)\lambda} \right]^{1/\beta} = \eta(1 - q)^{-1/\beta}, \quad (6)$$

so that $R(t) = 0$ for $t \geq T_{\max}$.

Remark 1. Asymmetric identifiability. As shown empirically in Section 6, the heavy-tail parameter $q > 1$ is identified by the tail decay rate and is estimable when the tail is observed, whereas for $q < 1$ the parameters $(q, \beta, T_{\max})$ are strongly coupled over the observed range and q is only weakly identifiable. This asymmetry motivates both the simulation design and the Bayesian regularisation developed below.

3 | Competing risks model

Consider K mutually exclusive failure causes with latent cause-specific lifetimes $T_1, \dots, T_K$, independently distributed with cause-specific survival functions $R_j(t)$. The observed data on each unit are

$$T = \min_{1 \leq j \leq K} T_j, \quad C = \arg \min_{1 \leq j \leq K} T_j, \quad (7)$$

with right-censoring indicated when the unit has not failed by the censoring time.

3.1 | Cause-specific likelihood

For a unit failing from cause j at time t , the density is $f_j(t) \prod_{\ell \neq j} R_\ell(t)$: cause j occurs in $(t, t+dt)$ while all other causes survive beyond t . A unit right-censored at t contributes $\prod_{j=1}^K R_j(t)$. Hence for data (t_i, c_i) , with $c_i = j$ for a cause- j failure and $c_i = 0$ (sentinel) for censoring, the likelihood is

$$\mathcal{L} = \prod_{i: c_i=j} f_j(t_i) \prod_{\ell \neq j} R_\ell(t_i) \prod_{i: \text{censored}} \prod_{j=1}^K R_j(t_i). \quad (8)$$

Taking logarithms, the log-likelihood *separates by cause*. The contribution of cause j is

$$\ell_j(\theta_j) = \sum_{i: c_i=j} \log f_j(t_i) + \sum_{i: c_i \neq j} \log R_j(t_i), \quad (9)$$

where the second sum includes both failures from other causes and censored units (for which cause j is right-censored). Thus $\log \mathcal{L} = \sum_{j=1}^K \ell_j$, and each cause may be fitted using a single-distribution routine in which its own failures are events and all other observations are right-censored.

3.2 | System survival and cumulative incidence functions

The system survival and system density are

$$R_{\text{sys}}(t) = \prod_{j=1}^K R_j(t), \quad f_{\text{sys}}(t) = \sum_{j=1}^K f_j(t) \prod_{\ell \neq j} R_\ell(t). \quad (10)$$

The cumulative incidence function (CIF) for cause j , the probability of failure from cause j by time t in the presence of all other causes, is

$$\text{CIF}_j(t) = \int_0^t f_j(s) \prod_{\ell \neq j} R_\ell(s) ds. \quad (11)$$

The CIFs partition the failure probability:

$$\sum_{j=1}^K \text{CIF}_j(t) = 1 - R_{\text{sys}}(t). \quad (12)$$

The eventual (crude) probability of cause j is $\text{CIF}_j(\infty)$. In the numerical work (11) is evaluated by trapezoidal integration on a fine time grid, and identity (12) provides a numerical check. To confirm numerical stability, the CIFs and system reliability were recomputed on grids of 500, 1000, 5000, and 10 000 points: the posterior summaries changed by less than 10^{-3} once the grid reached 1000 points, and all results below use a 5000-point grid.

4 | Bayesian formulation

4.1 | Parameterisation and priors

Direct sampling in (λ, β) is ill-conditioned because $\log \lambda = -\beta \log \eta$ couples the scale and shape, and the bounded regime adds

a hard support constraint. We instead work in the unconstrained, better-conditioned coordinates

$$\mathbf{x} = (x_q, x_e, x_b), \quad q = 2 - e^{x_q}, \quad \eta = e^{x_e}, \quad \beta = e^{x_b}, \quad (13)$$

with $\lambda = \eta^{-\beta}$ derived. The transformation for q is monotone from $\mathbb{R}$ to $(-\infty, 2)$, and those for η, β to $(0, \infty)$, so no constrained sampling is required.

Independent weakly informative priors are placed on the natural parameters

$$q \sim \mathcal{N}(\mu_q, \sigma_q^2), \quad \log \eta \sim \mathcal{N}(\mu_e, \sigma_e^2), \quad \log \beta \sim \mathcal{N}(\mu_b, \sigma_b^2), \quad (14)$$

with the q prior truncated to the admissible range. We use a data-driven default: $\mu_e = \log(\text{median event time})$, $\sigma_e = 1.5$; $\mu_b = \log 1.5$, $\sigma_b = 1.2$; and $\mu_q = 1$ (the ordinary Weibull as the regularising centre) with $\sigma_q = 0.5$ for competing-risks analysis. The scale priors are deliberately broad so that, when the data are informative, the likelihood dominates; the q prior's role is to regularise the limited-information tail parameter (Section 6).

We deliberately *do not* claim that $(q, \log \eta, \log \beta)$ is orthogonal. A natural alternative is $(q, \log \lambda, \log \beta)$, in which $\log \lambda = -\beta \log \eta$ might be expected to decouple scale from shape. We tested both parameterisations quantitatively on the voltage posterior, at the posterior mode, by the Hessian condition number $\kappa(\mathbf{H})$, the effective samples per second, the number of divergences, mean tree depth, and $\hat{R}$ (Figure 1, Table 1). The proposed coordinates give $\kappa \approx 1.75 \times 10^3$, 129.5 effective samples per second, 18 divergent transitions, mean tree depth 1.94, and $\hat{R}_{\text{max}} = 1.004$, whereas $(q, \log \lambda, \log \beta)$ gives $\kappa \approx 9.66 \times 10^5$ (about 550 times worse), 0.061 effective samples per second, 26 divergences, mean tree depth 5.74, and $\hat{R}_{\text{max}} = 1.17$, which fails to converge. We therefore find *no evidence* that $(q, \log \lambda, \log \beta)$ is the more orthogonal parameterisation; it is strictly worse, because $\log \lambda$ remains strongly coupled to β through $\log \lambda = -\beta \log \eta$. We keep $(q, \log \eta, \log \beta)$ and describe it as an *unconstrained parameterisation with substantially reduced global posterior anisotropy*. We are explicit that this is a relative, not absolute, improvement: even in our chosen coordinates a strong pairwise correlation remains, $|\rho_{\text{max}}| \approx 0.94$ between $\log \eta_D$ and $\log \beta_D$ (the degradation-mode scale and shape), and $\kappa \approx 1.75 \times 10^3$ is far from unit condition. What the reparameterisation buys us is a much smaller *global* condition number and effective-sample throughput; the residual within-mode correlation is absorbed by the Hessian preconditioner of Section 5.2. The divergence counts in Table 1 are from this shorter comparison run (two chains $\times$ 600 draws); the production voltage fit (two chains $\times$ 2000 draws, 1000 warm-up) is diagnosed separately in Section 5.4.

TABLE 1 | Parameterisation comparison on the voltage posterior at the mode: A = $(q, \log \eta, \log \beta)$, B = $(q, \log \lambda, \log \beta)$.

Quantity	A	B
Hessian condition number $\kappa(\mathbf{H})$	1.75×10^3	9.66×10^5
Effective samples / second	129.5	0.061
Divergent transitions	18	26
Mean tree depth	1.94	5.74
$\hat{R}_{\text{max}}$	1.004	1.17

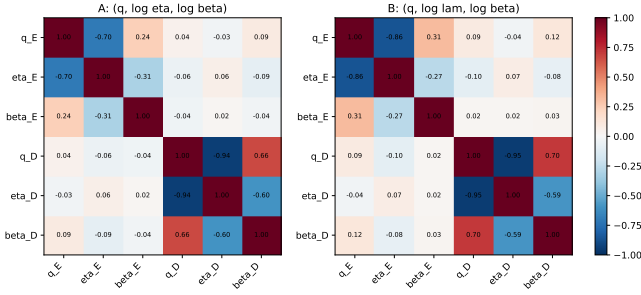


FIGURE 1 | Posterior correlation structure (parametric comparison). The alternative $(q, \log \lambda, \log \beta)$ (B) is not more orthogonal: its Hessian is about $550\times$ worse-conditioned and its chains fail to converge. We retain $(q, \log \eta, \log \beta)$ as a better-conditioned, unconstrained system.

4.2 | Posterior and analytic gradient

Because the priors in (14) are expressed in the natural coordinates while sampling is in $\mathbf{x}$, the change of variables for $q = 2 - e^{x_q}$ contributes the Jacobian term $\log |\partial q / \partial x_q| = x_q$; the priors on $\log \eta, \log \beta$ are already on x_e, x_b . The log-posterior is

$$\log p(\mathbf{x} | \text{data}) = \ell(\mathbf{x}) - \frac{1}{2} \left(\frac{q - \mu_q}{\sigma_q} \right)^2 + x_q - \frac{1}{2} \left(\frac{x_e - \mu_e}{\sigma_e} \right)^2 - \frac{1}{2} \left(\frac{x_b - \mu_b}{\sigma_b} \right)^2. \quad (15)$$

Let g_q, g_λ, g_β denote the derivatives of the log-likelihood with respect to the raw parameters (q, λ, β) , obtained analytically. Chaining through (13) and $\lambda = \eta^{-\beta}$,

$$\frac{\partial \ell}{\partial x_e} = -\beta \lambda g_\lambda, \quad \frac{\partial \ell}{\partial x_b} = \beta (g_\beta - g_\lambda \lambda x_e), \quad \frac{\partial \ell}{\partial x_q} = -e^{x_q} g_q. \quad (16)$$

The likelihood is differentiable at $q = 1$. Writing $z = \lambda t^\beta$, the limiting q -derivative is

$$g_q|_{q=1} = \sum_{\text{events}} \left(\frac{z^2}{2} - 1 \right) + \sum_{\text{censored}} \left(z + \frac{z^2}{2} \right). \quad (17)$$

The analytic gradient (16) agrees with a numerical gradient to relative error below 10^{-8} , and the cause-separable form (9) lets the competing-risks posterior reuse the single-cause routine for each mode.

5 | NUTS computation

5.1 | Hamiltonian dynamics and the No-U-Turn Sampler

HMC augments the state with an auxiliary momentum $\mathbf{r}$ and defines the Hamiltonian

$$H(\mathbf{x}, \mathbf{r}) = -\log p(\mathbf{x}) + \frac{1}{2} \mathbf{r}^\top \mathbf{M}^{-1} \mathbf{r}, \quad (18)$$

where $\mathbf{M}$ is a mass (preconditioning) matrix. The canonical equations are simulated by the leapfrog integrator, which alternates half-kicks in momentum and full steps in position. The No-U-Turn Sampler builds a balanced binary trajectory by recursively doubling the leapfrog path in a randomly chosen direction, stopping when

Algorithm 1 NUTS for q-Weibull (competing risks), with Hessian mass and dual-averaging step size

Require: data (t_i, c_i) ; initial point $\mathbf{x}_0$; target acceptance δ ; draws N , warmup W

- 1: $\mathbf{M} \leftarrow \text{regularize}(-\nabla^2 \log p(\mathbf{x}_0))$; $\mathbf{M}^{-1}$; $\mathbf{L} \leftarrow \text{chol}(\mathbf{M})$
- 2: choose step size ε by the find-reasonable-epsilon routine
- 3: **for** $m = 1$ **to** $N + W$ **do**
- 4: draw $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$; $\log u \leftarrow \log p(\mathbf{x}) - \frac{1}{2} \mathbf{r}^\top \mathbf{M}^{-1} \mathbf{r} + \log U$
- 5: recursively double the trajectory (both directions) until a U-turn, divergence, or max depth; track valid points
- 6: set $\mathbf{x}$ to a slice-weighted valid point; flag a divergence if the energy error exceeds a threshold
- 7: **if** $m \leq W$ **then**
- 8: adapt ε by primal-dual averaging toward δ
- 9: **end if**
- 10: **end for**
- 11: **return** post-warmup draws and diagnostics ($\hat{R}$, ESS, divergences, acceptance, tree depth)

the path doubles back on itself (a U-turn), when a divergence is detected, or at a maximum tree depth; a slice variable provides a valid, detailed-balance-preserving proposal over the valid points, as implemented in Stan [8]. Algorithm 1 summarises the procedure.

5.2 | Mass matrix and step-size adaptation

Rather than treating any matrix as the exact local precision, we use a local Hessian-based preconditioner to reduce posterior anisotropy. We form $\mathbf{M}$ from the negative Hessian of the log-posterior at the mode, symmetrise it, and regularise its spectrum so that no direction is too steep or too flat (Section B); we then draw $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$ and use $\mathbf{M}^{-1}$ in the leapfrog position update. By convention the Hamiltonian kinetic energy is $K(\mathbf{r}) = \frac{1}{2} \mathbf{r}^\top \mathbf{M}^{-1} \mathbf{r}$, so that the momentum covariance is exactly $\mathbf{M}$. This is a practical preconditioner, not an assertion that the local Hessian equals the true posterior covariance. The step size is initialised with the find-reasonable-epsilon routine, which doubles or halves it while a fixed initial momentum gives a single-step acceptance probability respectively below or above one half. It is then tuned during warmup by primal-dual averaging toward a target acceptance of 0.85--0.90. The step size is additionally capped at $0.9 \times 2 / \sqrt{\lambda_{\max}(\mathbf{M})}$, the leapfrog stability limit. Re-estimation of the mass matrix from warmup draws is available but not required, as the initial Hessian preconditioner is already effective.

5.3 | Smooth barrier for the bounded regime

For $q < 1$ the support endpoint (6) can lie close to the largest observation, where the log-density has a singular curvature. A Hamiltonian step that crosses this hard wall is rejected with infinite energy, and the chain can become stationary while the step-size adaptation collapses to zero. We replace the hard logarithm of the inner factor

(2) by a smooth extension. For a small threshold $\varepsilon_s = 10^{-3}$,

$$\begin{aligned}\widetilde{\log}(u) &= \begin{cases} \log u, & u > \varepsilon_s, \\ \log \varepsilon_s + (u - \varepsilon_s)/\varepsilon_s, & u \leq \varepsilon_s, \end{cases} \\ \widetilde{\log}'(u) &= \begin{cases} 1/u, & u > \varepsilon_s, \\ 1/\varepsilon_s, & u \leq \varepsilon_s. \end{cases}\end{aligned}\quad (19)$$

Inside the data-supported region $u > \varepsilon_s$ the density is unchanged. We are explicit that we sample a *computationally smoothed target*, not the exact singular posterior: this smoothed target agrees *pointwise* with the original likelihood throughout the entire data-supported region. At the fitted parameters the smallest inner factor over the observed units is well above the threshold (about 0.49 in the bounded-regime design below, and 0.84--1.01 at the voltage posterior mean), so the ramp is never evaluated at a data point and the likelihood on the observed sample is identical, not merely close. Beyond the nominal boundary the log-density continues as a finite-slope ramp off which trajectories can reflect rather than stick. The barrier is therefore *computationally active but inferentially inactive*: it acts only in the unobserved near-wall region that the trajectories explore, and a sensitivity analysis in Section 6.3 confirms that posterior summaries are unchanged for ε_s from 10^{-3} to 10^{-5} .

5.4 | Convergence diagnostics

We run multiple chains from dispersed initial points and report, following [23]: the potential-scale-reduction factor $\hat{R}$ for each parameter (target $\hat{R} < 1.01$); the bulk effective sample size (ESS); the number of divergent transitions, separated for warmup and the sampling phase; the mean acceptance probability; and the mean tree depth. Initial points with non-finite log-posterior (e.g. a random perturbation crossing the support boundary) are detected and re-drawn with decreasing perturbation scale, falling back to the maximum-likelihood point, preventing a chain from initialising outside the support.

5.4.1 | Where the divergent transitions occur.

For the production voltage fit (two chains $\times$ 2000 draws after 1000 warm-up draws, target acceptance 0.9), divergent transitions are separated into warm-up (98) and post-warm-up (95). Their location is instructive (Table 2): about 93 of the 95 post-warm-up divergences ($\approx 98\%$) occur where the sampler explores the artificial smooth-barrier or support-wall region--- q pinned against the soft walls at $q \approx 0.2$ or $q \approx 1.98$, or a $q < 1$ trajectory briefly penetrating the support boundary---and only 2 of 95 land inside the data-supported posterior bulk (Mahalanobis distance 1.4--1.8). The remaining diagnostics are healthy: maximum post-warm-up tree depth 4 (mean 2.06), mean acceptance 0.933, per-parameter ESS [494, 486, 615, 776, 800, 964] (total 4135), about 68.3 effective samples per second, per-chain BFMI [1.023, 1.007] (mean 1.015), and $\hat{R} < 1.01$. At the posterior mean the smallest inner factor is $u = 1.013$ (early mode) and 0.838 (degradation mode), both far above $\varepsilon_s = 10^{-3}$. The honest reading is that *about 98%* of post-warm-up divergences occur in the barrier/support-wall region rather than in the high-density bulk; the

small residual fraction (2 of 95) means we do not claim the sampler never touches the bulk. With BFMI ≈ 1.01 , ESS/s ≈ 68 , and $\hat{R} < 1.01$, these barrier-localised divergences do not materially affect the posterior summaries.

TABLE 2 | Divergence localisation and convergence diagnostics for the production voltage NUTS fit (2 chains $\times$ 2000 draws, 1000 warm-up, target acceptance 0.9).

Quantity	Value
Warm-up divergences	98
Post-warm-up divergences	95
near barrier / support wall	≈ 93 ($\approx 98\%$)
in posterior bulk (Mahalanobis 1.4--1.8)	2
Tree depth, mean / max	2.06 / 4
Acceptance probability	0.933
Per-parameter ESS	[494, 486, 615, 776, 800, 964]
Total ESS	4135
Effective samples / second	68.3
BFMI (mean per chain)	[1.023, 1.007] (1.015)
$\hat{R}_{\max}$	< 1.01
Min inner factor (E / D)	1.013 / 0.838

6 | Simulation study

6.1 | Design

The simulation examines estimation of q across the bounded, ordinary-Weibull, and heavy-tailed regimes, crossed with three levels of competing truncation. The cause of interest is a q -Weibull with median lifetime $E(T) = 100$ and shape $\beta = 2$, evaluated at $q \in \{0.6, 0.8, 1.0, 1.25, 1.45, 1.7\}$: $q < 1$ gives a bounded support, $q = 1$ the ordinary Weibull, and $q > 1$ a heavy tail. The competing cause is an ordinary Weibull with $\beta = 3$, whose scale is set from the truncation ratio $\rho = \text{median}(T_c)/\text{median}(T)$, with $\rho \in \{0.6, 1.0, 4.0\}$ producing respectively *strong*, *moderate*, and *weak* truncation of the target cause. For each of the 18 conditions, $n = 200$ units are generated from (7), and the experiment is replicated $R = 80$ times.

Three estimates of the target q are compared: (i) the **MLE** of the correctly specified q -Weibull, obtained by BFGS, with a 95% Wald interval from the numerical information; (ii) the **Bayesian** estimate under the default weakly informative prior $q \sim \mathcal{N}(1, 0.5^2)$, sampled by NUTS with 200 warm-up and 300 post-warm-up draws, using the posterior median and the equal-tailed 95% credible interval; and (iii) the **Weibull MLE** with q fixed at 1, a misspecified baseline. We report bias, RMSE, and 95% interval coverage of q . The Weibull baseline returns $q = 1$ identically, with bias and RMSE equal to $|q - 1|$ and zero coverage whenever $q \neq 1$, which quantifies the cost of omitting the shape parameter.

6.2 | Results

Table 3 and Figure 3 summarise the results.

Under *strong truncation*, the competing Weibull removes most units before the tail of the target cause is observed. Because q is identified through tail decay, limited tail information produces a broad, asymmetric likelihood and a systematic downward bias, as the partially observed tail resembles a lighter distribution. Both estimators

TABLE 3 | Estimation of q across six true values and three truncation levels ($R = 80, n = 200$): bias, RMSE, and 95% interval coverage for the q-Weibull MLE and the Bayesian (NUTS) estimates; the smaller RMSE is bold. The fixed-Weibull baseline is described in the text.

q	Truncation	MLE			Bayes (NUTS)		
		Bias	RMSE	Coverage	Bias	RMSE	Coverage
0.6	Strong	0.06	0.60	0.71	0.39	0.42	0.99
	Moderate	0.01	0.39	0.89	0.22	0.29	0.99
	Weak	-0.07	0.21	0.95	-0.00	0.17	0.71
0.8	Strong	-0.06	0.61	0.76	0.22	0.27	1.00
	Moderate	-0.06	0.44	0.91	0.08	0.21	1.00
	Weak	-0.06	0.18	0.97	-0.03	0.15	0.70
1.0	Strong	-0.22	0.66	0.80	0.02	0.17	1.00
	Moderate	-0.16	0.48	0.89	-0.05	0.22	1.00
	Weak	-0.04	0.14	0.94	-0.04	0.13	0.78
1.25	Strong	-0.28	0.68	0.80	-0.18	0.24	1.00
	Moderate	-0.15	0.45	0.91	-0.16	0.28	0.95
	Weak	-0.03	0.10	0.96	-0.04	0.11	0.82
1.45	Strong	-0.28	0.66	0.81	-0.31	0.37	0.96
	Moderate	-0.07	0.32	0.95	-0.20	0.28	0.91
	Weak	-0.02	0.07	0.96	-0.04	0.09	0.95
1.7	Strong	-0.11	0.38	0.88	-0.35	0.42	0.82
	Moderate	-0.02	0.13	0.96	-0.14	0.22	0.89
	Weak	-0.01	0.06	0.94	-0.04	0.08	0.89

share this truncation bias. Bayesian shrinkage toward the regularising centre nevertheless lowers the RMSE for every q except 1.7 (ratios 0.26--0.71, Figure 3a), and restores near-nominal coverage from the under-covering MLE intervals (0.71--0.88 to 0.96--1.00). The gain is largest when q is close to the centre of the prior ($q = 1$, ratio 0.26) and shrinks as q moves away.

Under *moderate truncation* the profile likelihood combines a local peak near the truth with broad shoulders (Figure 5). The Bayesian estimate retains the advantage for $q \leq 1.25$ (ratios 0.45--0.74), is close to the MLE at $q = 1.45$ (0.88), and is worse at $q = 1.7$ (1.72), where the prior pulls against a value the partial data still locate. Under *weak truncation* the tail is largely observed and q is well identified, so the likelihood dominates: the two estimates are close for $q \leq 1$, whereas for $q \geq 1.25$ the MLE is better (ratios 1.06--1.34), because shrinkage adds a small bias without a variance reduction.

The extremes delimit the useful range of the regularising prior. At $q = 1.7$, far from the prior centre, Bayesian shrinkage does not improve on the MLE under any truncation (1.09, 1.72, 1.34). In the bounded regime the credible intervals are conservative under strong and moderate truncation (coverage 0.99--1.00) but under-cover under weak truncation (0.70--0.78), indicating that the bounded tail is harder to calibrate than the heavy tail. Taken together, the results map the region in which Bayesian regularisation helps: when information is scarce and q departs only moderately from the ordinary Weibull, it reduces RMSE and restores coverage, whereas when the data locate q well or q lies far from the centre, the MLE is preferable.

6.2.1 | Where the point-estimate gain comes from

A referee rightly asked whether the RMSE improvement is a Bayesian phenomenon or merely the effect of regularising q . To separate these, on a representative subset ($q \in \{0.8, 1.0, 1.45\}$, strong and weak truncation, $n = 200, R = 40$, prior $q \sim \mathcal{N}(1, 0.5^2)$) we computed, for each replicate, the unpenalised MLE, the *penalised*

MLE (maximisation of log-likelihood plus the same quadratic q penalty), the posterior mode (MAP), and the full posterior median and mean. Table 4 reports the bias and RMSE of each q estimator.

Under *weak truncation* all five estimates agree to within 10--20% (e.g. $q = 1.45$: MLE RMSE 0.08, penalised 0.09, MAP 0.09, posterior median 0.09, mean 0.09): with the tail observed, regularisation adds nothing. Under *strong truncation* the unpenalised MLE is badly biased and variable (RMSE 0.61, 0.67, 0.71 for $q = 0.8, 1.0, 1.45$), whereas the penalised MLE already recovers most of the gain (0.32, 0.24, 0.43) and the posterior median is close to it (0.27, 0.18, 0.39). The honest reading is that, in the weakly identified (strongly truncated) region, the dominant point-estimate improvement comes from the quadratic q penalty / prior shrinkage as *regularisation*, not from Bayesian computation per se: the penalised MLE reproduces most of the RMSE reduction. The added value of the full Bayesian analysis is the calibrated posterior interval (coverage, Section 6.2) that a penalised point estimate cannot supply; on point estimation alone the penalised MLE and the Bayes posterior are nearly interchangeable.

6.2.2 | Observed information and the weak-identifiability map

We can tie the loss of identifiability directly to the observed information. On the $(q, \log \eta, \log \beta)$ scale we evaluated the expected (Fisher) information matrix $\mathbf{I}$ at the design values, its determinant, its smallest eigenvalue, and its condition number (Figure 2). Moving from strong to weak truncation, $\det(\mathbf{I})$ grows by about two orders of magnitude---from $\approx 3.3 \times 10^4$ (strong, $q = 0.6$) to $\approx 8.9 \times 10^6$ (weak, $q = 1.45$)---while the smallest eigenvalue rises from ≈ 3 to ≈ 14 --33; the condition number stays in the range 33--132. The reason is the fraction of units failing from the target cause, $\text{frac}_{\text{event}}$: it is only 0.28--0.37 under strong truncation, about 0.54 under moderate, and 0.82--0.97 under weak truncation. The $\text{RMSE}(q)$ of the MLE tracks exactly this information scarcity, falling from 0.6--0.7 (strong) to below 0.1 (weak). Figure 2 colour-codes each true q , with

TABLE 4 | Decomposition of the regularisation gain for q (bias, RMSE; $n = 200$, $R = 40$, prior $q \sim \mathcal{N}(1, 0.5^2)$). Entries are **bias** (RMSE). Under weak truncation all five estimators agree; under strong truncation the unpenalised MLE is poor and the quadratic penalty (penalised MLE or MAP) already recovers most of the RMSE reduction.

q_{true}	Truncation	Estimator of q (bias, RMSE)				
		MLE	Penalised MLE	MAP	Bayes median	Bayes mean
0.8	Strong	−0.16 (0.61)	+0.20 (0.32)	+0.05 (0.19)	+0.20 (0.27)	+0.16 (0.23)
0.8	Weak	−0.05 (0.15)	−0.02 (0.13)	−0.04 (0.13)	−0.02 (0.14)	−0.03 (0.14)
1.0	Strong	−0.33 (0.67)	+0.01 (0.24)	−0.15 (0.23)	−0.01 (0.18)	−0.05 (0.17)
1.0	Weak	−0.04 (0.13)	−0.04 (0.12)	−0.06 (0.12)	−0.04 (0.11)	−0.04 (0.12)
1.45	Strong	−0.38 (0.71)	−0.33 (0.43)	−0.49 (0.54)	−0.32 (0.39)	−0.37 (0.42)
1.45	Weak	−0.03 (0.08)	−0.04 (0.09)	−0.06 (0.09)	−0.05 (0.09)	−0.06 (0.09)

open-circle solid lines for the MLE and filled-circle dashed lines for the Bayes RMSE.

Because $\det(\mathbf{I})$ depends on the units of the parameterisation, it is only an auxiliary summary. A more interpretable measure of the information *specifically about* q , after the nuisance scale and shape ($\log \eta$, $\log \beta$) are eliminated, is the Schur complement

$$I_{q \cdot \psi} = I_{qq} - I_{q\psi} \mathbf{I}_{\psi\psi}^{-1} I_{\psi q}, \quad \text{SE}(q) \approx I_{q \cdot \psi}^{-1/2}, \quad (20)$$

together with the width W_q of the 95% profile-likelihood (likelihood-ratio) interval for q . Table 5 reports these across truncation and true q . Under strong truncation the partial information about q is small ($I_{q \cdot \psi} = 5.2$ –48.6), the standard error is large (0.14–0.44), and W_q is wide—and, for the bounded and near-Weibull cases, *one-sided*: the lower profile-LOD bound hits the q -grid edge ($q = 0.25$), so the interval cannot close on the left. As truncation weakens, $I_{q \cdot \psi}$ rises to 115 and the interval shrinks to $W_q \approx 0.23$ (weak, $q = 1.45$). This scale-free picture is the quantitative content of “weak identifiability”: strong competing truncation leaves little information about the tail shape once the nuisance parameters are profiled out, and the profile interval is not merely wide but occasionally unbounded on one side.

TABLE 5 | Scale-free identifiability of q : Schur-complement partial information $I_{q \cdot \psi}$, implied standard error, and 95% profile-LR interval width W_q ($n = 200$). An asterisk marks a one-sided interval whose lower end hits the q -grid boundary ($q = 0.25$).

Truncation	q_{true}	$I_{q \cdot \psi}$	$\text{SE}(q)$	W_q
Strong	0.6	5.2	0.44	1.45*
	1.0	12.6	0.28	1.47*
	1.45	48.6	0.14	1.39
Moderate	0.6	11.4	0.30	1.37*
	1.0	24.6	0.20	1.06
	1.45	80.7	0.11	0.43
Weak	0.6	21.0	0.22	0.99*
	1.0	38.7	0.16	0.61
	1.45	115.1	0.09	0.23

6.2.3 | Recovery of the scale and shape η, β

Table 6 reports the recovery of η and β in addition to q ($R = 30$, Monte Carlo standard error ≈ 0.04 ; treated as a descriptive, exploratory supplement rather than a formal Monte Carlo validation). Under weak truncation all three parameters are recovered cleanly: the η bias is single-digit and its RMSE 10–13 (on the 100-unit

median-life scale), and the β bias is small throughout. Under *strong* truncation, by contrast, η becomes severely distorted—for $q = 1.45$ strong the MLE η RMSE is ≈ 87 —but the Bayesian interval is correspondingly wide and still covers the truth (≈ 0.9). The β estimates remain stable. One further, regime-specific pathology appears for the bounded $q = 0.6$ under strong truncation: the support endpoint T_{max} is occasionally estimated as infinite, i.e. the posterior escapes to $q \geq 1$, because the bounded mode is essentially unidentifiable when almost no units fail from it. This is consistent with the conservative, wide bounded-regime intervals noted earlier.

TABLE 6 | Bayesian recovery of q, η, β (bias, RMSE, 95% coverage; $n = 200$, $R = 30$). η is severely distorted only under strong truncation, where the wide interval still covers the truth.

Condition	Par.	Bias	RMSE	Coverage
$q = 1.45$ weak	q	−0.05	0.09	0.90
	η	+6.2	13.5	0.93
	β	−0.07	0.20	0.97
$q = 1.45$ strong	q	−0.31	0.39	0.93
	η	+38.5	50.6	0.90
	β	−0.11	0.31	0.90
$q = 0.6$ weak	q	−0.01	0.15	0.77
	η	−1.1	9.7	0.77
	β	−0.00	0.15	0.77

The computational comparison (Figure 4) benchmarks four samplers on a representative strongly truncated data set ($q = 1.45$): random-walk Metropolis, adaptive Metropolis, a preconditioned HMC with a fixed trajectory length, and NUTS. When the random walk is given the exact posterior covariance estimated from the MLE Hessian, a choice that deliberately favours it, its wall-clock efficiency (92 effective samples per second) matches NUTS (90). In this six-dimensional, well-preconditioned setting NUTS therefore does not gain in effective samples per second.

Its advantage lies elsewhere. First, NUTS adapts its mass matrix during warm-up and needs no externally supplied covariance, whereas the competitive Metropolis result depends on that precomputed input. Second, the fixed-trajectory HMC over-integrates (acceptance 1.00) and obtains only 0.091 effective samples per gradient evaluation, about half the 0.179 attained by NUTS, which selects the trajectory length automatically. NUTS also remains reliable from dispersed starts and as the dimension or anisotropy grows, where a single fixed-covariance proposal deteriorates.

As an external sanity check, we fit the ordinary-Weibull ($q = 1$) competing-risks baseline on the voltage data with a mature implementation (PyMC 6.3.2, ArviZ 1.3.0; dense-mass NUTS with

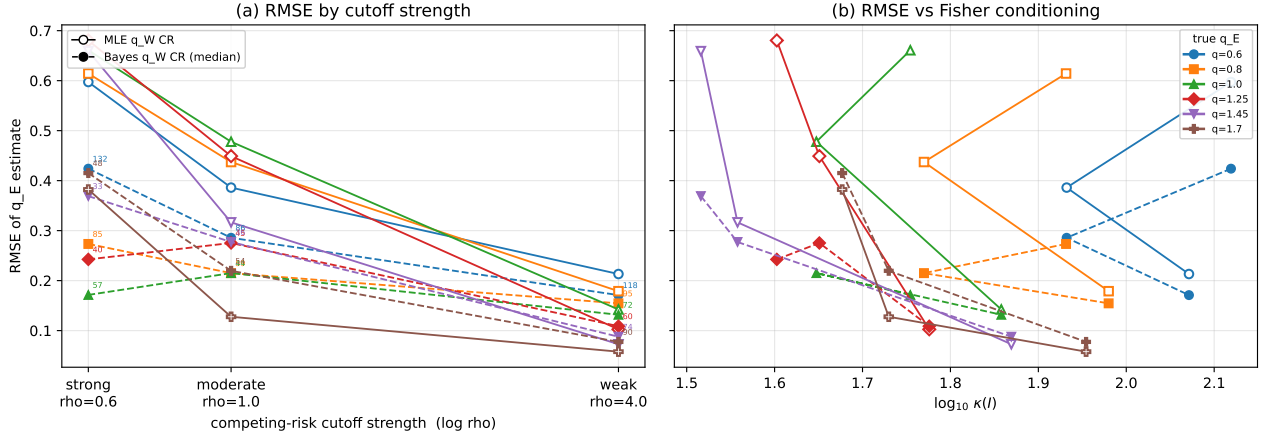


FIGURE 2 | Weak-identifiability map. Observed information ($\det \mathbf{I}$, $\lambda_{\min}$, κ) and the target-cause event fraction rise monotonically as truncation weakens, and MLE $\text{RMSE}(q)$ tracks this information scarcity. Open circles/solid: MLE; filled circles/dashed: Bayes.

dual averaging): it gives zero divergent transitions, $\hat{R}_{\max} = 1.002$, bulk effective sample size 1382, and about 81 effective samples per second on pure sampling (the 77 s wall time includes a one-time just-in-time compilation). Our own NUTS, on the strictly harder six-parameter q -Weibull posterior, attains about 129 effective samples per second with $\hat{R}_{\max} \approx 1.004$, i.e. throughput of the same order as a specialised library on a smaller, easier model.

6.3 | Sensitivity analyses

We first vary the prior on q , reusing the same strongly truncated data sets ($q = 1.45$, $R = 40$). The default $\mathcal{N}(1, 0.5^2)$ gives the best balance (RMSE 0.39, coverage 0.95). A tighter $\mathcal{N}(1, 0.25^2)$ under-covers (0.72) with a narrow interval, a wider $\mathcal{N}(1, 1^2)$ raises the RMSE to 0.49 but still improves on the MLE (0.66), showing that the gain is not solely the effect of shrinkage, and a flat $\mathcal{U}(-1, 2)$ gives an unstable point estimate (RMSE 1.05) although its wide interval still covers the truth (0.95). With no regularisation the weakly identified posterior is dominated by the boundary. On the voltage data the early-mode posterior becomes harder to sample as the prior widens (more divergent transitions), yet under every prior the 95% interval for q_E contains 1; the absence of strong heavy-tail evidence therefore does not rely on a prior centred at the Weibull value.

The boundary threshold is assessed on the bounded $q = 0.6$, strong-truncation design ($R = 40$, Table 7). Posterior summaries are identical for $\varepsilon_s = 10^{-3}, 10^{-4}, 10^{-5}$ (bias 0.38, RMSE 0.42, coverage 0.97, interval width 1.47), and $\varepsilon_s = 10^{-2}$ differs only marginally. The smallest inner factor at the fitted parameters is about 0.49, so the ramp is inactive at every data point; the threshold changes only the frequency of divergent transitions in the unobserved near-wall region, not the estimates.

TABLE 7 | Sensitivity to the smooth-barrier threshold ε_s on the bounded $q = 0.6$, strong-truncation design ($R = 40$).

ε_s	Bias	RMSE	Coverage	Interval width	Divergences
10^{-2}	0.39	0.44	0.97	1.49	0.12
10^{-3}	0.38	0.42	0.97	1.47	3.02
10^{-4}	0.38	0.42	0.97	1.47	3.40
10^{-5}	0.38	0.42	0.97	1.47	3.40

6.4 | Prior centre μ_q

Because the default prior is centred at the ordinary Weibull value ($\mu_q = 1$), we checked how the regularisation target affects the strongly truncated, heavy-tailed case ($q_{\text{true}} = 1.45$, $n = 200$, $R = 30$, $\sigma_q = 0.5$). Centring at $\mu_q = 0.8$ gives bias -0.479 , RMSE 0.534, coverage 0.90; the default $\mu_q = 1.0$ gives -0.311 , 0.388, 0.93; and $\mu_q = 1.2$ gives -0.207 , 0.302, 0.97. The bias and RMSE improve monotonically as the centre approaches the truth, and the coverage rises to near-nominal. This confirms that the remaining downward shrinkage bias under strong truncation (about -0.31 at the default centre) is a property of anchoring at $q = 1$, not a sampling artefact; in practice the centre should be placed near the physically expected tail behaviour whenever prior information exists.

6.5 | Upper-truncation boundary fraction.

For the bounded regime the location of the observational window relative to the support endpoint matters. Define $f = \max_i t_i / T_{\max}$, the fraction of the theoretical endpoint reached by the largest observed time ($q = 0.6$, weak truncation, $n = 200$, $R = 20$; descriptive/-exploratory). At $f = 0.5$ (about 35% censoring) the MLE biases q upward by $+0.13$ and the Bayes estimate by $+0.28$, with coverage falling to 0.65; at $f = 0.7$ the bias is $+0.05$ and coverage 0.85; for $f \geq 0.85$ the estimates return to the weak-truncation baseline (bias ≈ 0 , coverage ≈ 0.75). Heavy upper truncation against a bounded support thus pushes the bounded q upward and degrades coverage, because the finite endpoint is essentially unobserved.

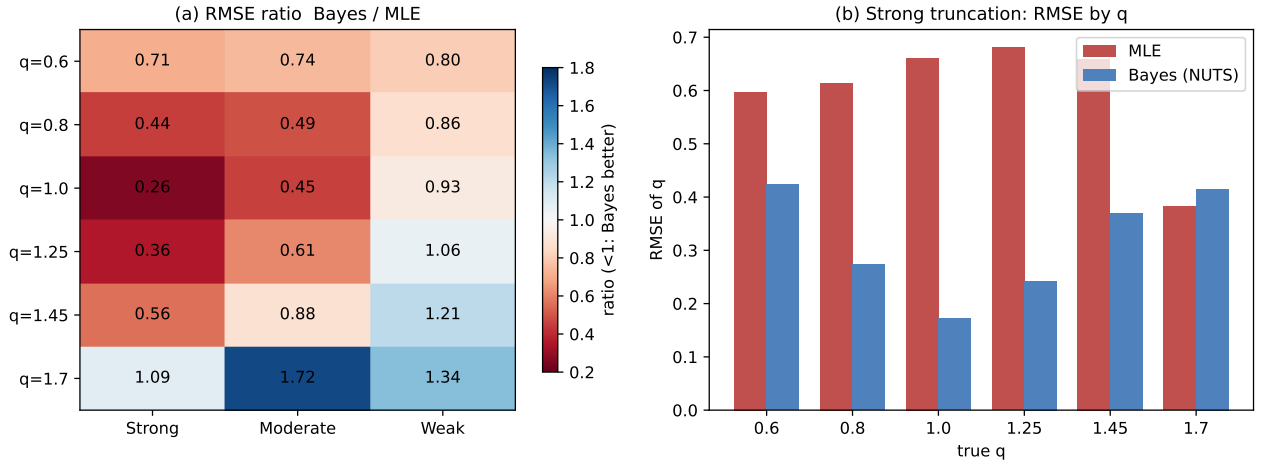


FIGURE 3 | (a) RMSE ratio Bayes/MLE across q and truncation (red: Bayes better, blue: MLE better); (b) RMSE under strong truncation. Bayesian regularisation helps mainly under strong truncation with moderate departures from $q = 1$.

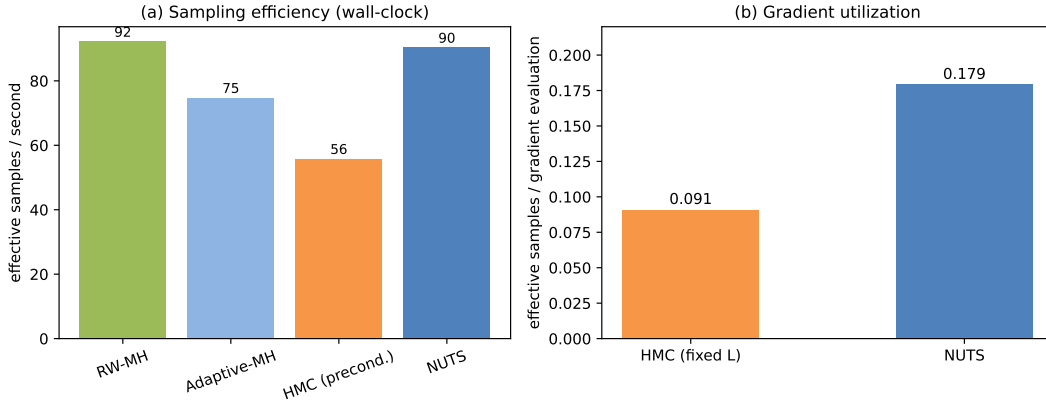


FIGURE 4 | (a) Effective samples per second for four samplers; a covariance-preconditioned random walk matches NUTS when given the exact covariance. (b) Effective samples per gradient evaluation; NUTS uses gradients about twice as efficiently as a fixed-trajectory HMC.

6.6 | Sample size.

Finally, Figure 6 varies n under the hardest design ($q_{\text{true}} = 1.45$, strong truncation). The MLE RMSE stays on a weak-identification plateau---0.73, 0.78, 0.76 for $n = 50, 100, 200$ ---and collapses only at $n = 500$ to 0.20; the Bayesian RMSE is flatter and better-calibrated (0.37, 0.39, 0.40, 0.26) with coverage at least 0.92 throughout. This shows that, under strong competing truncation, increasing the sample size in the hundreds does not by itself restore identifiability: the regularisation and the honest interval are what stabilise inference until the information (sample size and/or observed tail) becomes large. The effective information is governed by $n_{\text{eff}} \approx n \times p_{\text{tail}}$, where p_{tail} is the fraction of units that survive long enough to inform the tail: under strong truncation p_{tail} is small, so even $n = 200$ carries little tail information. The $n = 500$ point uses only $R = 12$ replicates (Monte Carlo standard error ≈ 0.063), so this sample-size scan is treated as *exploratory* rather than as a formal Monte Carlo validation. Figure 6 colour-codes the four curves with distinct markers (MLE/Bayes RMSE and |bias|).

7 | Application to high-voltage stress-test data

7.1 | Data description

We analyse the `voltage` data from a high-voltage stress test of generator armature bars [11], distributed with the `weibulltools` package [21]. Fifty-eight independent units were followed until failure or censoring; 18 failed from an *early* mode (E), 27 from a *degradation* mode (D), and 13 were right-censored. Figure 7 shows the event times: E failures are concentrated at short times while D failures occur later, consistent with two distinct mechanisms.

7.2 | Model comparison and posterior inference

Table 8 compares a single Weibull, a Weibull competing-risks model, and a q-Weibull competing-risks model by maximum likelihood.

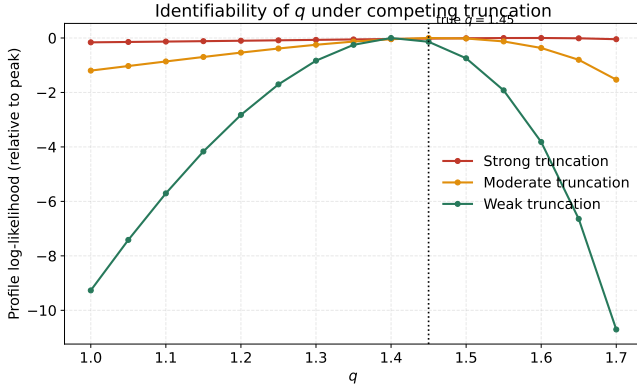


FIGURE 5 | Profile log-likelihood of q under strong, moderate, and weak competing truncation. Early competing failure broadens the profile and weakens identifiability; with weak truncation the profile sharpens at the true value.

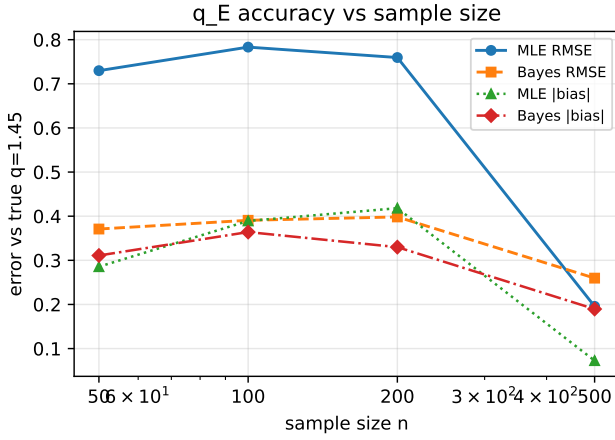


FIGURE 6 | Estimation error versus sample size under strong truncation ($q = 1.45$). The MLE RMSE stays on a weak-identification plateau for $n = 50$ – 200 and drops only at $n = 500$; the Bayesian RMSE is flatter and its coverage stays at least 0.92.

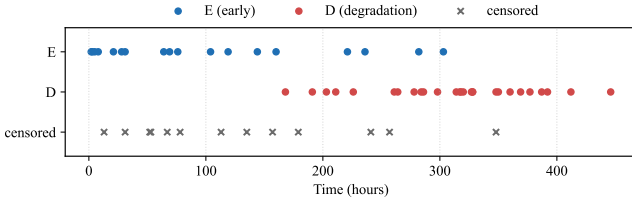


FIGURE 7 | Event and censoring times by failure mode for the voltage data.

Allowing two causes is strongly supported: the likelihood-ratio statistic against the single Weibull is 10.92 ($p \approx 0.004$). Adding q , however, is not significant ($2\Delta\ell = 0.038$, $p \approx 0.98$), and the q-Weibull model is penalised by AIC [4] and BIC [20]. The Bayesian posterior (Table 9, Figure 8) tells a consistent and nuanced story. The early mode shows a weak heavy-tail signal: the posterior is shifted above one with $\Pr(q_E > 1) \approx 0.76$, but its interval includes one, whereas the degradation mode is centred at $q_D \approx 0.97$ and is adequately represented by an ordinary Weibull. We report this uncertainty transparently rather than claim heavy tails the data do

TABLE 8 | Maximum-likelihood comparison for the voltage data.

Model	No. params	Log-lik.	AIC	BIC
Single Weibull	2	−292.53	589.06	593.18
Weibull CR	4	−287.07	582.13	590.37
q-Weibull CR	6	−287.05	586.10	598.46

not establish. For the early mode the characteristic life η_E is also only weakly determined (95% interval 91–2429 h), because the scale is confounded with the heavy-tail shape over the short early-mode window; the degradation-mode scale η_D is estimated much more precisely (297–409 h).

TABLE 9 | Bayesian posterior summaries for the q-Weibull competing-risks model (NUTS, target acceptance 0.90).

Parameter	Median	95% interval	$\Pr(q > 1)$
q_E	1.23	[0.64, 1.71]	0.76
η_E (h)	599	[91, 2429]	--
β_E	0.67	[0.44, 0.98]	--
q_D	0.97	[0.51, 1.38]	0.46
η_D (h)	350	[297, 409]	--
β_D	5.23	[3.61, 7.60]	--

7.3 | Bayesian Weibull baseline.

Restricting both causes to ordinary Weibulls in the Bayesian fit ($q = 1$, four parameters) gives a regularised comparison: $\eta_E = 925$ h (95% interval 474–2788), $\beta_E = 0.68$ (0.43–0.98), $\eta_D = 345$ h (320–376), and $\beta_D = 5.31$ (3.80–6.92), with no divergent transitions and posterior log-likelihood median −289.1. The q-Weibull Bayesian fit reaches no higher posterior log-likelihood, consistent with the likelihood-ratio result ($2\Delta\ell = 0.038$). Freeing q therefore adds no predictive value on these data, and the Weibull fit is both more parsimonious and numerically cleaner. The framework can represent a heavy tail when that tail is observed, as the simulation shows, but here it correctly withholds that interpretation.

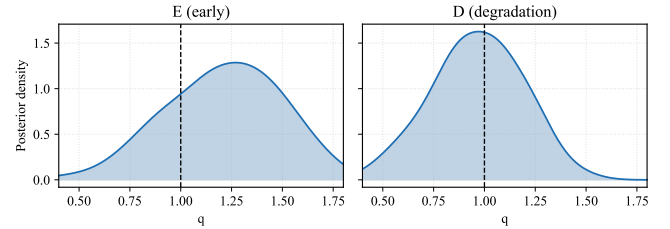


FIGURE 8 | Posterior density of q for the early and degradation modes; the dashed line marks $q = 1$ (ordinary Weibull).

7.4 | Cumulative incidence and cause probabilities

Figure 9 shows the posterior system reliability and CIFs. The estimated eventual cause probabilities are 0.37 for E (95% interval 0.26–0.48) and 0.63 for D (0.52–0.74), close to the observed failure proportions ($18/45 = 40\%$, $27/45 = 60\%$). At the largest observed time (446 h), the system reliability is 0.018, with $\text{CIF}_E = 0.37$ and $\text{CIF}_D = 0.62$. For every posterior draw the CIFs and system reliability satisfy $\text{CIF}_E(t) + \text{CIF}_D(t) + R_{\text{sys}}(t) = 1$ up to a numerical

integration error below 0.01; we report posterior means, which remain additive, rather than separately computed medians. The fitted system reliability closely tracks the non-parametric Kaplan–Meier estimate [16], supporting the parametric model.

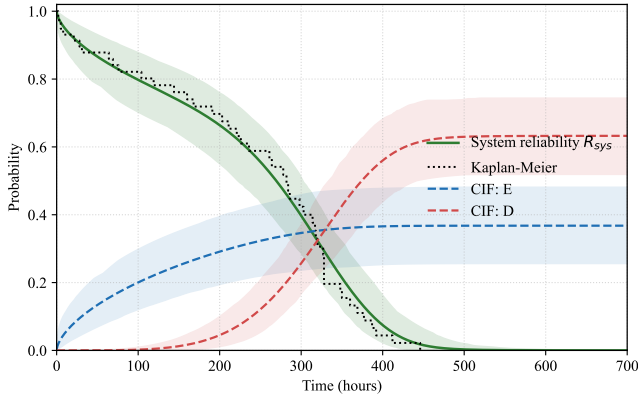


FIGURE 9 | Posterior system reliability and cause-specific cumulative incidence functions with uncertainty bands, with Kaplan–Meier comparison.

7.5 | Posterior predictive check

To verify that the fitted model can reproduce the salient features of the data, we drew $B = 300$ posterior replicates (re-applying the observed censoring times) and compared observed statistics to their posterior predictive distribution (Figure 10). The observed 18 early-mode failures sit inside the 95% predictive interval [10, 28.5] (median 19), the 27 degradation-mode failures inside [21, 38] (median 29), and the largest observed lifetime 446 h inside [390, 549] h (median 442 h). The cumulative incidences at the last observed time agree: $\text{CIF}_E(446) = 0.346$ versus predictive [0.197, 0.549], and $\text{CIF}_D(446) = 0.617$ versus [0.431, 0.799]. The modelled system reliability over 50–400 h lies entirely within the 95% predictive band. No discrepancy was detected for the selected predictive summaries: the q-Weibull competing-risks model reproduces the failure counts, the range of lifetimes, and the cause-specific accumulation; the PPC therefore does not reveal a structural misspecification, while it remains consistent with the earlier conclusion that the heavy-tail signal for the early mode is weak.

8 | Discussion

The findings clarify both the value and the limits of Bayesian inference for q-Weibull competing risks. The central mechanism is identifiability: the entropic parameter q is read mainly from the tail, and a competing cause that truncates the tail removes exactly the information needed to estimate it. The resulting likelihood is broad and asymmetric, and its shape differs by regime. Heavy-tailed $q > 1$ is identifiable once the tail is observed, whereas bounded $q < 1$ is intrinsically harder because q , β , and the support endpoint are partially confounded over any finite observation window.

Bayesian regularisation is beneficial precisely in the limited-information regime: a weakly informative prior centred at the ordinary Weibull reduces estimator variance (and RMSE) without

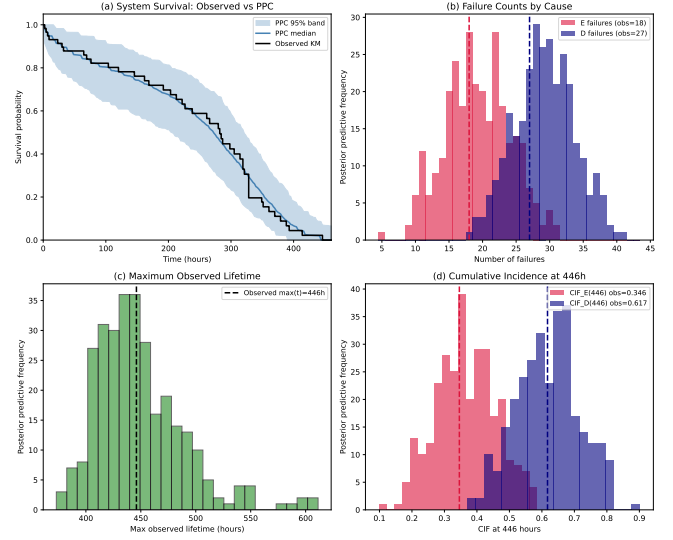


FIGURE 10 | Posterior predictive check for the voltage data ($B = 300$ draws). Observed summaries (vertical dashed lines) lie inside their 95% predictive intervals (shaded bands / histograms) for the failure counts, the maximum lifetime, the cause-specific CIFs at 446 h, and the system reliability over 50–400 h.

claiming to remove the systematic truncation bias, and the full posterior communicates uncertainty that a Wald interval cannot. When the data are informative, the same broad prior recedes and the likelihood dominates, so the method does not impose shrinkage where it is unwarranted. We caution against interpreting a posterior shifted above one as proof of heavy tails without an interval that excludes one, as the voltage application illustrates.

On the computational side, NUTS with an analytic gradient, Hessian preconditioning, and the smooth barrier gives reliable, high-ESS inference even at the bounded support wall where naive samplers freeze.

A key limitation is the assumption of independent latent cause-specific times. To gauge its effect we simulated data from a shared-gamma-frailty model (so the two causes are dependent) but fitted the *wrong* independent q-Weibull competing-risks Bayes model ($q_{\text{true}} = 1.45$, strong truncation, $n = 200$, $R = 20$; exploratory). With independent data the bias in q_E is -0.30 (RMSE 0.39, coverage 0.90); under mild dependence (frailty variance $\nu = 5$) it shrinks to -0.08 (RMSE 0.23, coverage 1.00), i.e. mild dependence has only a limited effect; but under stronger dependence ($\nu = 2$) it flips to $+0.25$ (RMSE 0.31, coverage 0.80), which substantially distorts the q_E inference. Ignoring positive frailty dependence thus moves the q_E bias from negative to positive and degrades coverage to 0.80: the independent-cause assumption is acceptable under weak dependence but should be relaxed, e.g. with shared-frailty or copula-cause models, when dependence is suspected.

The hardest region of all is the bounded regime $q < 1$ when the support endpoint is not observed. There the profile interval can be one-sided and the Bayesian coverage can fall to about 0.65 (Section on boundary fraction), and the posterior occasionally “escapes” to $q \geq 1$ because the bounded mode is essentially unidentifiable. Regularisation stabilises inference in this region but *cannot create information about an unobserved support boundary*; no amount of prior centring recovers an endpoint that the data never approach. Other

limitations include the choice of prior scale and the modest sample size of the application. Natural extensions are accelerated life tests with stress-dependent scale parameters, hierarchical and covariate-dependent models, dependent causes, and robust inference under contamination [6]; a recent survey of q-Weibull reliability applications is given by [24]. We also refer to [13] for the general Bayesian workflow.

9 | Conclusion

We developed a unified Bayesian framework for q-Weibull competing risks, computed by NUTS in a better-conditioned, unconstrained coordinate system $(q, \log \eta, \log \beta)$ with an analytic gradient and a smooth barrier for the bounded support wall. Across the bounded, Weibull, and heavy-tailed regimes, the simulation showed that competing truncation weakens the identifiability of the tail parameter q , and it mapped where Bayesian regularisation helps: it can substantially improve point estimation and interval calibration under strong competing-risk truncation, but it is *not* a universal improvement over the MLE—the likelihood dominates when the tail is observed or q lies far from the prior centre. The core qualification is that regularisation stabilises inference but cannot create information about an unobserved support boundary. On the computational side, NUTS adapts its mass matrix automatically and uses gradient evaluations about twice as efficiently as a fixed-trajectory HMC; it matches a covariance-preconditioned random walk only in the special case where that walk is handed the exact posterior covariance. Applied to high-voltage stress-test data, the method established significant competing risks, identified a weak and uncertain heavy-tail signal in the early mode, and confirmed an ordinary Weibull for the degradation mode, reporting the non-significance of q without overstatement. The framework provides a practical and honest basis for q-Weibull competing-risks analysis and a foundation for accelerated-test and hierarchical extensions.

Author Contributions

Huirong Cao: conceptualisation, methodology, software, formal analysis, visualisation, writing—original draft. Fuchang Wang: supervision, methodology, writing—review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability

The voltage data are publicly available through the `weibulltools` R package [21]. All simulation, fitting, plotting, and result-generation code, together with the machine-readable result files used

in this paper, is released as a reproducible research package at <https://github.com/fzmth/qweibull-cr> (a README documents the directory layout and build).

APPENDIX

A | Details of the No-U-Turn sampler

This appendix supplies the implementation details summarised in Algorithm 1. We use the standard NUTS construction of [14, 8]. Given a current point $\mathbf{x}$ and a momentum draw $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$, a slice variable $u \sim \text{Uniform}(0, \exp[-H(\mathbf{x}, \mathbf{r})])$ defines the set of admissible points. The tree is built by recursion: at each doubling the leapfrog trajectory is extended by one more step in the current direction from either the leftmost or the rightmost leaf, and the proposal is chosen uniformly among the leaves whose joint density exceeds u . The recursion stops when (i) the two outermost leaves double back on themselves (U-turn, $(\mathbf{x}^+ - \mathbf{x}^-)^\top \mathbf{r} < 0$), (ii) a leapfrog step produces a divergent transition (the squared Hamiltonian jump exceeds the divergence threshold, calibrated from the energy distribution), or (iii) a maximum tree depth (we use 10) is reached. Divergent transitions are rejected for the target estimate but retained for diagnostics.

During warmup, the step size ε is tuned by primal-dual averaging [14] toward a target acceptance probability $\delta = 0.85\text{--}0.90$: a dual variable adapts $\log \varepsilon$ with a decreasing gain, and the running average is what is finally used, which stabilises the adaptation. The mass matrix $\mathbf{M}$ is set once from the Hessian preconditioner below; re-estimation from the first half of warmup draws is implemented but found unnecessary here.

B | Analytic gradient and Hessian preconditioner

The log-posterior gradient used in the leapfrog steps is the analytic expression of Section 4.2. For each cause, the raw likelihood gradients $(g_q, g_\lambda, g_\beta)$ of Equations (3)–(4) are obtained by differentiating (9) term by term, using the inner factor $u(t) = 1 - (1 - q)\lambda t^\beta$ and its derivatives; the $q = 1$ limit (17) removes the apparent $0/0$ singularity. The cause-separable form (9) means the six-parameter gradient is the concatenation of two three-cause gradients. These analytic derivatives agree with central finite differences to relative error below 10^{-8} .

The preconditioning mass matrix is built from the negative Hessian $\mathbf{H} = -\nabla^2 \log p(\mathbf{x})$ evaluated at the posterior mode (or, before fitting, at the MLE). Because the raw Hessian can be indefinite or ill-conditioned in the weakly identified directions, we regularise it in one of two equivalent ways: (i) add a small ridge, $\mathbf{M} = \mathbf{H} + \varepsilon \mathbf{I}$ with $\varepsilon = 10^{-3} \text{tr}(\mathbf{H})/d$, and (ii) an eigenvalue clip, $\lambda_i \leftarrow \max(\lambda_i, \lambda_{\max}/\kappa_{\max})$ with $\kappa_{\max} = 10^4$, which prevents any direction from being too flat (which would inflate the leapfrog step) or too steep (which would shrink it). The draw $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$ and the position update $\mathbf{x} \leftarrow \mathbf{x} + \varepsilon \mathbf{M}^{-1} \mathbf{r}$ then use the symmetrised, regularised factorisation. This is a local preconditioner to reduce posterior anisotropy, not an exact posterior covariance (Section 5.2).

C | Reproducing the build

The manuscript is compiled with XeLaTeX against the Wiley NJD USG class:

```
xelatex -interaction=nonstopmode main.tex
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xelatex -interaction=nonstopmode main.tex
xelatex -interaction=nonstopmode main.tex
```

Python 3.10+ with numpy, scipy, pandas, matplotlib, and (for the external baseline only) pymc 6.x / arviz 1.x reproduces all numbers and figures; the result JSONs in data/ are the exact sources of every table entry above.

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Huirong Cao is an Associate Professor in the School of Science at Langfang Normal University, Langfang, China. Her research interests include reliability statistics, Bayesian computation, computational actuarial science, and risk management.

Fuchang Wang is a Professor at the University of Emergency Management, Sanhe, China. His research interests include statistical computing, machine learning, and artificial intelligence.